

Requirements of transmitter and receiver in WDM networks, types of tunable lasers, optical amplifiers, SOA and FOA, EDFA and EBFA.



WDM (Wavelength Division Multiplexing)

- In **WDM (Wavelength Division Multiplexing)** networks, multiple data signals are sent simultaneously over a single optical fiber by using different wavelengths (colors) of laser light.
- To make this work, the hardware—transmitters, receivers, and amplifiers—must meet specific high-performance requirements.

1. Transmitter and Receiver Requirements

- **Transmitter Requirements:**

- **Precision Wavelength:** The laser must emit **a very specific wavelength to avoid interfering with adjacent channels** (Crosstalk).
- **Stability:** The wavelength **must remain constant regardless of temperature changes** (often using "wavelength lockers").
- **Narrow Linewidth:** The light beam must be **"pure" so the signal doesn't spread out (dispersion) over long distances.**

wavelength lockers

- Wavelength lockers হলো অপটিক্যাল কমিউনিকেশন সিস্টেমে ব্যবহৃত একটি গুরুত্বপূর্ণ ডিভাইস, যা লেজারের নির্দিষ্ট তরঙ্গদৈর্ঘ্য (wavelength) স্থির রাখতে সাহায্য করে।
- লেজার থেকে যে আলো বের হয়, তার wavelength কখনো কখনো তাপমাত্রা বা অন্য কারণে পরিবর্তিত হতে পারে। কিন্তু **fiber optic communication**-এ নির্দিষ্ট wavelength খুব গুরুত্বপূর্ণ (বিশেষ করে WDM সিস্টেমে)।
 - এই wavelength ঠিক রাখার কাজই করে wavelength locker।

- **Receiver Requirements:**

- **Selectivity:** The receiver must be able to pick out one specific wavelength from the many arriving at the end of the fiber.
- **Sensitivity:** It must be able to detect very weak signals after they have traveled hundreds of kilometers.
- **Wide Bandwidth:** It needs to handle the high data rates (e.g., 100Gbps or 400Gbps) sent by the transmitter.

2. Types of Tunable Lasers

- A tunable laser can change its operating wavelength.
- This is useful because one backup laser can replace any failed laser in a WDM system.
 - **External Cavity Lasers (ECL):** Uses a physical grating that moves to change the color. Very precise but mechanically complex.
 - **Distributed Bragg Reflector (DBR):** Uses electrical current to change the refractive index of a built-in grating, changing the wavelength.
 - **Vertical Cavity Surface Emitting Lasers (VCSEL):** Small and cheap; they tune by moving a tiny top mirror using MEMS technology.

MEMS technology

- MEMS technology এর পূর্ণরূপ হলো Micro-Electro-Mechanical Systems।
- MEMS হলো এমন একটি প্রযুক্তি যেখানে খুব ছোট (micro-level) **mechanical + electrical components** একসাথে একটি চিপের মধ্যে তৈরি করা হয়।
- **উদাহরণ:**
 - Mobile phone-এর **accelerometer** (ফোন ঘোরালে screen rotate হয়)
 - Car airbag sensor
 - Inkjet printer head

3. Optical Amplifiers

- In the past, to boost a signal, we had to convert light to electricity, amplify it, and turn it back to light.
- Modern **Optical Amplifiers** boost the light signal directly.

3. Optical Amplifiers

• SOA (Semiconductor Optical Amplifier)

- **Mechanism:** Uses a semiconductor chip (similar to a laser but without the mirrors).
- **Pros:** Small, can be integrated on circuits.
- **Cons:** Higher noise and lower power than fiber-based amplifiers.

• EDFA (Erbium-Doped Fiber Amplifier)

- **Mechanism:** A piece of glass fiber is "doped" with Erbium ions.
 - A "pump laser" excites these ions, and when the signal passes through, the ions release energy to boost the signal.
- **Standard:** This is the "gold standard" for long-distance internet because it can amplify many WDM channels at once without distortion.

• EBFA (Erbium-Bismuth Fiber Amplifier)

- **Evolution:** Similar to EDFA but uses Bismuth-doped glass to provide a wider bandwidth, allowing even more WDM channels to be packed into the fiber.

Example: The Highway Analogy

- Imagine a 40-lane highway (WDM Fiber).
 - **Transmitter:** A specialized ramp that ensures **only Blue cars enter Lane 1 and only Red cars enter Lane 2.**
 - **Receiver:** A specialized exit that **only lets the Blue cars off at a specific destination.**
 - **Amplifier (EDFA):** A "speed boost" zone in the middle of the highway that **gives every car more fuel as they drive through, without them having to stop at a gas station.**

WDM Power Calculation

- A WDM system has 8 channels. Each channel enters an EDFA with a power of -20 dBm. The EDFA (Erbium-Doped Fiber Amplifier) has a gain of 25 dB.

- Calculate the output power of a single channel in dBm.
- Calculate the total output power of the EDFA in milliwatts (mW).

Solution:

1. Output per channel: $P_{out(dBm)} = P_{in(dBm)} + Gain_{(dB)}$

$$P_{out} = -20 \text{ dBm} + 25 \text{ dB} = 5 \text{ dBm}$$

2. Total Power in mW:

First, convert 5 dBm to mW: $P_{(mW)} = 10^{(P_{dBm}/10)} = 10^{(5/10)} = 3.16 \text{ mW}$.

Since there are 8 identical channels:

$$TotalPower = 8 \times 3.16 \text{ mW} = 25.28 \text{ mW}.$$